

Your Name

Mechatronics Engineering Student

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Projects

Autonomous Line-Following Robot

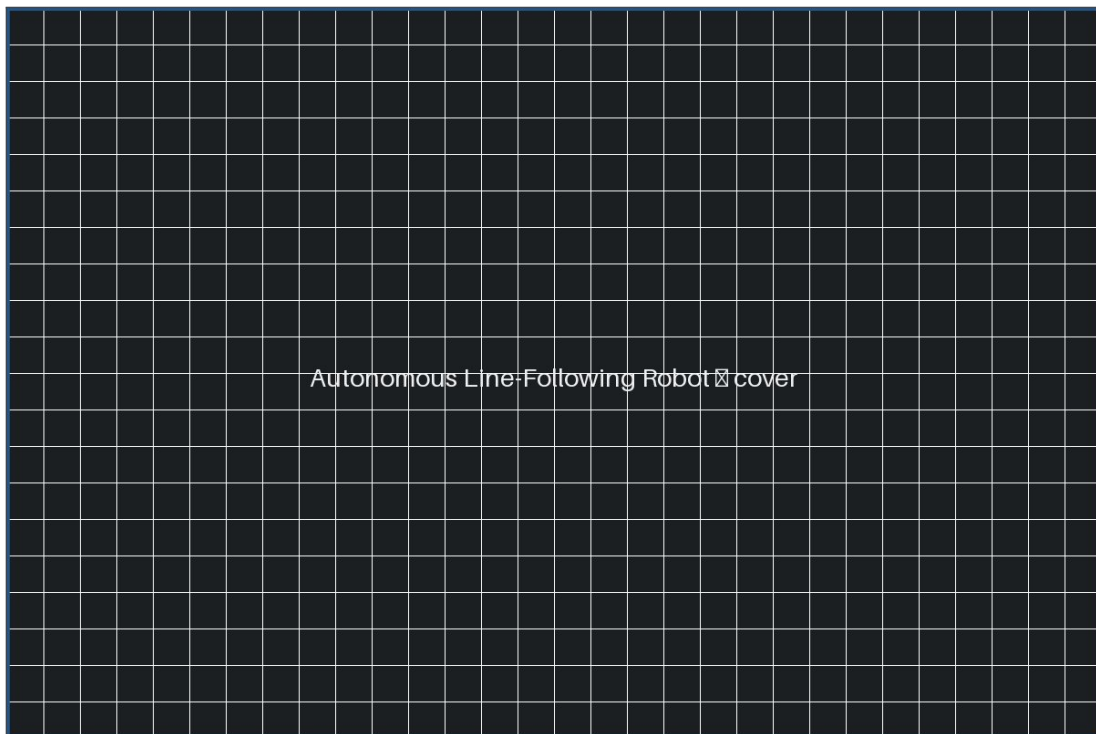
PID-tuned mobile robot with adaptive speed control

DATES Jan 2026 – Mar 2026

TAGS Embedded Systems, Control Theory, PCB Design

TOOLS STM32, KiCad, C, MATLAB

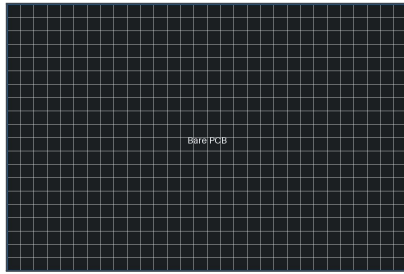
A two-wheeled autonomous robot that follows a taped track using an IR sensor array and a tuned PID control loop, holding speed through sharp corners without losing the line.



Final chassis, top view

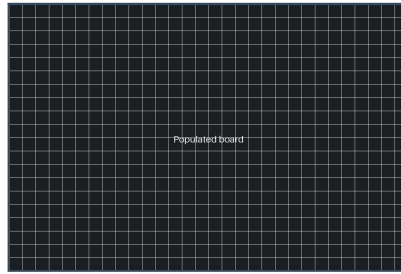
Custom PCB mounted above the IR sensor array and drive motors.

Build process



Bare PCB

First revision board fresh off the mill.



Populated board

Sensor array and motor drivers soldered in.

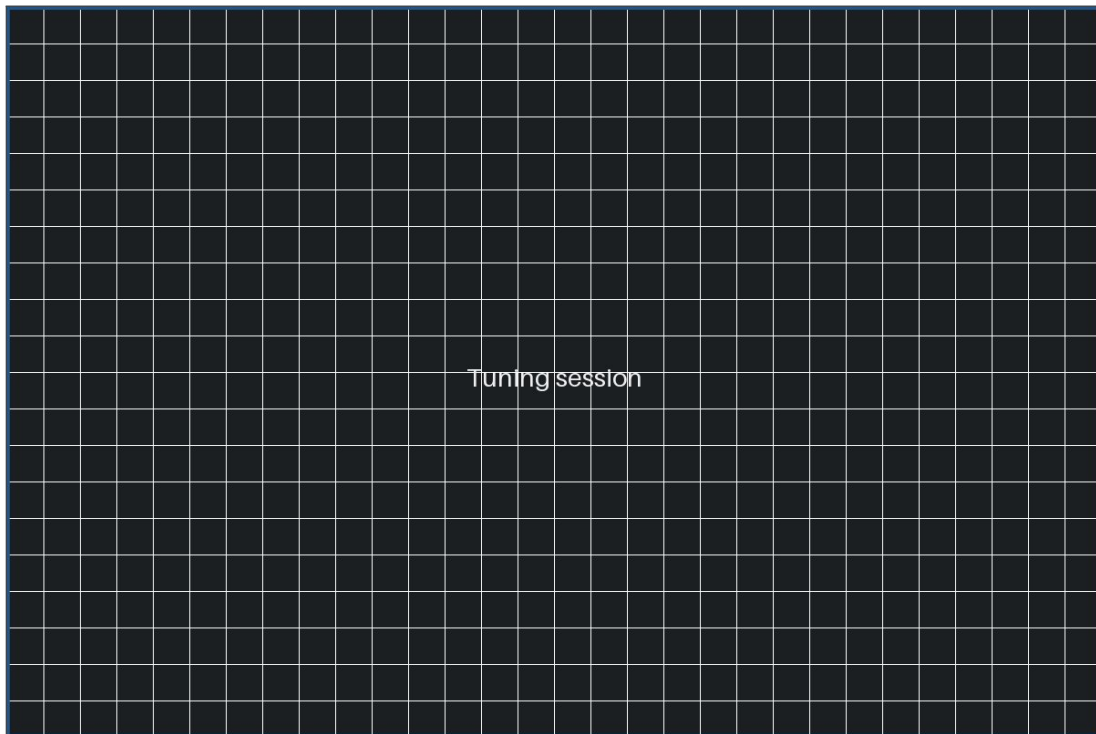


Chassis assembly

3D-printed chassis with motors mounted.

Control loop redesign

The first revision used a fixed-gain proportional controller, which oscillated badly on tight corners. Moving to a full PID loop with gain scheduling based on error magnitude let the robot hold near-top speed on straights while slowing smoothly into turns.



Tuning session

Logging sensor error against wheel speed to hand-tune PID gains.

Final lap times improved by 38% over the proportional-only version, with zero track departures across 20 consecutive test laps.

Modular PCB Sensor Hub

Stackable environmental sensor boards over I2C

DATES May 2025 – Jul 2025

TAGS PCB Design, Firmware, IoT
TOOLS Altium Designer, STM32, I2C

A stackable sensor board system where each board adds one environmental sensor over a shared I2C bus, letting a project mix and match temperature, humidity, and air-quality sensing without redesigning the base board.

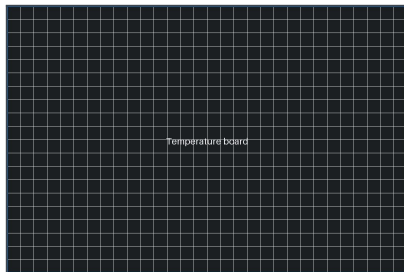


Three-board stack

Base board with humidity and air-quality daughter boards attached.

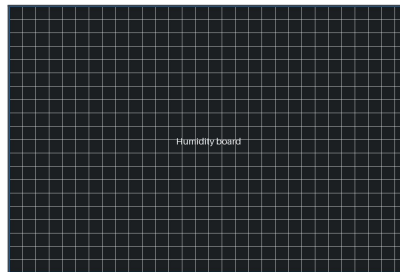
Each daughter board carries a unique I2C address set by a resistor-strapped address pin, so boards can be stacked in any order and combination without firmware changes on the base.

Individual boards



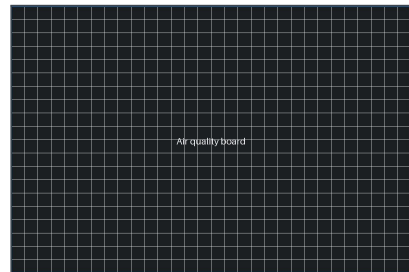
Temperature board

Single-sensor board, smallest form factor in the stack.



Humidity board

Includes onboard heater trace for condensation prevention.



Air quality board

VOC sensor with dedicated shielding to reduce PCB heat coupling.



Bus topology

Shared I2C bus routed through the board-to-board stacking connector.

3-DOF Desktop Robotic Arm

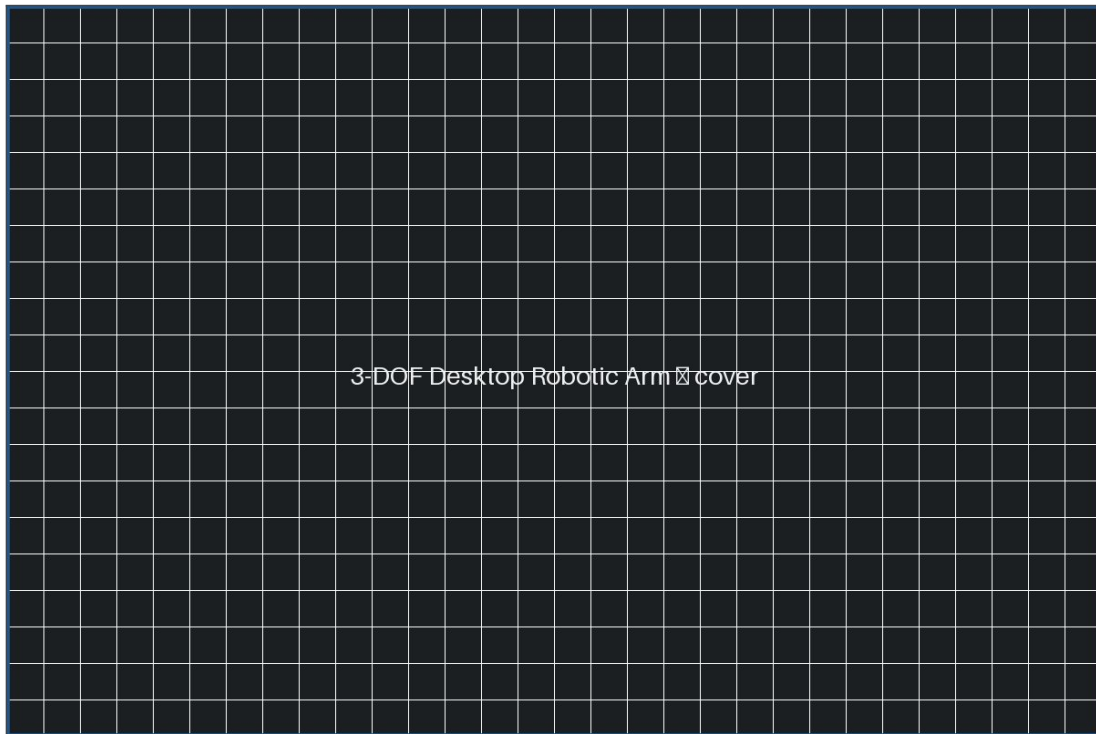
Inverse-kinematics-driven pick-and-place arm

DATES Sep 2025 – Dec 2025

TAGS Robotics, CAD, Kinematics

TOOLS SolidWorks, Arduino, Python

A three-degree-of-freedom robotic arm with a custom inverse kinematics solver, designed and 3D-printed from scratch for a desktop pick-and-place demo.



Assembled arm

Full assembly with gripper attachment at rest position.



CAD overview

Full linkage assembly modeled before any printing began.

Rather than use a lookup table, I derived closed-form inverse kinematics for the 3-link

planar arm so any target (x, y) within the workspace maps directly to joint angles — no iterative solving needed on the microcontroller.

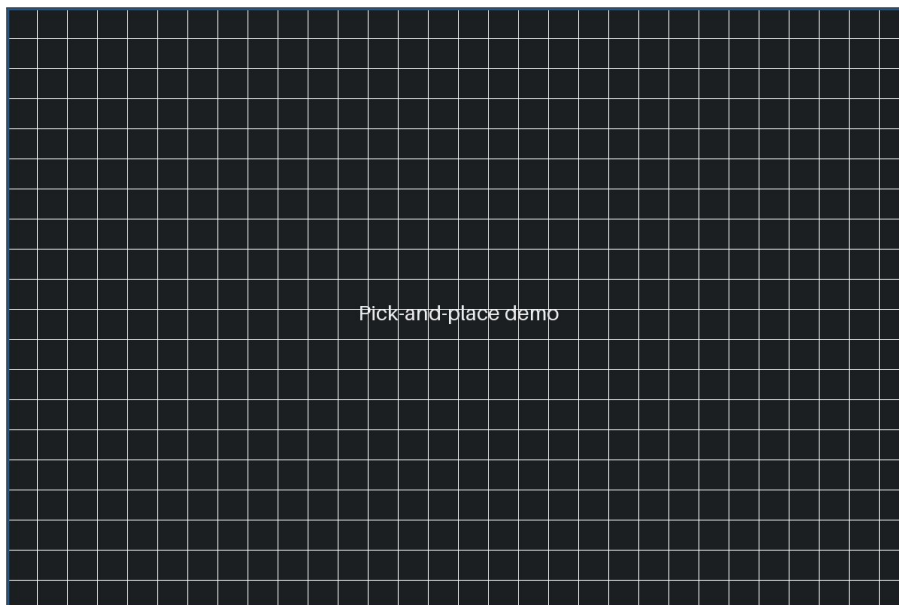


[\[3D MODEL --- VIEW INTERACTIVE ON SITE\]](#)

[Gripper jaw \(interactive\)](#)

Drag to rotate — this is the actual print file for the final jaw geometry.

The gripper jaw went through four print iterations to get enough compliance to grip irregular objects without a force sensor, using a living-hinge geometry at the jaw base.



[\[VIDEO --- VIEW INTERACTIVE ON SITE\]](#)

[Pick-and-place demo](#)

Arm sorting objects by color using a webcam and OpenCV.

Design Team Experience

Formula SAE — Powertrain Subteam

Electronics Lead

DATES Sep 2025 – Present

ORG University Formula SAE Team

Lead a 4-person electronics subteam responsible for the car's telemetry system, designing and maintaining the CAN bus network that connects the ECU, battery management system, and dash display.

Internships

Mechatronics Intern

Automation Engineering

DATES May 2025 – Aug 2025

ORG Example Robotics Co.

Designed and validated a vision-guided pick-and-place fixture for a production line, cutting cycle time by 12% and reducing manual changeover steps from 6 to 2.